

Aussie Primary Academy | Australian Curriculum v9.0 | Learn, Investigate and Explain

Name: _____

Date: _____

LEARNING INTENTION	examine how people use observations and data to develop and improve scientific explanations
KEY IDEA	Data are recorded observations or measurements. People look for patterns, compare repeated results and develop explanations that fit the evidence. New data can strengthen, limit or change an explanation.
WORKED MODEL	Repeated temperatures are lower beneath tree shade than on nearby concrete. The pattern supports an explanation about shade reducing heating, but comparable methods and more locations improve confidence.

SUCCESS CRITERIA - I CAN☐ identify data and patterns ☐ connect evidence to an explanation ☐ recognise limits ☐ revise a claim**A. BUILD THE SCIENCE IDEA**

1. Label observation, measurement, pattern or explanation in four statements about playground temperature.

2. Explain why a table of repeated readings is more useful than one memory.

3. Underline the evidence and circle the explanation in the worked model.

B. APPLY AND EXPLAIN

4. Data are 18, 20 and 19 C in shade; 27, 29 and 28 C in sun. Describe the pattern precisely.

5. Write an explanation using the pattern and an appropriate science idea.

6. State one claim these data cannot support.

C. INDEPENDENT CHECK

7. Create a claim-evidence-reasoning organiser for the data.

8. Revise this claim: Shade always makes every place exactly 9 C cooler.

Aussie Primary Academy | Australian Curriculum v9.0 | Application, Reasoning and Self-Review

Name: _____

Date: _____

D. APPLY SCIENCE IN CONTEXT

9. Two classes measure water loss from open and covered containers. Results differ by 2 mL between classes.

10. Compare their methods and results before deciding whether a common explanation is supported.

11. Write a cautious conclusion and identify one useful new measurement.

E. REASON, EVALUATE AND TRANSFER

12. Evaluate: If data fit my prediction, my explanation must be correct.

13. Show how one new result could strengthen, limit or change an explanation.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

[] I can use data and patterns to develop a scientific explanation.

Circle one: independently | with one reminder | with support

Evidence from my work that proves I understand: _____

The most difficult science idea was _____ because _____

One correction or improvement I made: _____

A question I still have: _____

Aussie Primary Academy | Australian Curriculum v9.0 | Teacher Copy - Answers and Teaching Support

Accept equivalent responses when the student uses accurate science ideas, relevant evidence and clear reasoning.

COMPLETE ANSWER KEY

1. Observation uses senses; measurement has number/unit; pattern summarises relationship; explanation gives a science reason.
2. Repeated recorded data can be checked and compared.
3. Temperatures are evidence; reduced heating is the explanation.
4. Shade results cluster at 18-20 C; sun results at 27-29 C, about 9 C higher each trial.
5. Example: shade blocks some incoming light energy, so the surface gains less heat.
6. It cannot establish all places, times, surfaces or causes from this small test.
7. Claim answers; evidence cites values; reasoning links the science idea.
8. Under these test conditions, shade was about 9 C cooler in three trials.
- 9-11. Compare container, volume, time, location and measuring equipment; small variation may still support a common pattern.
12. False; alternative explanations and method quality must be considered.
13. A repeated consistent result strengthens; an exception limits scope; a conflicting pattern may require revision.

TEACHER CHECK AND NEXT STEP

[] Uses accurate vocabulary and a complete cause-and-effect explanation.

[] Supports conclusions with observations or data.

Next step: revisit one misconception, then transfer the idea to a new example.

COMMON MISCONCEPTION / WATCH OUT

A pattern describes what the data show; an explanation tells why it may occur.

Matching a prediction does not prove that only one explanation is possible.

EXTENSION AND RE-TEACH

Re-teach: model one difficult item, complete one together, then ask the student to explain an independent correction.

Extension: apply the science idea to a new real-world example and justify the explanation with observations or data.